Methods for optimizing the training of Neural Networks

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Backpropagation Recap: Error Propagation

The forward step:

$$z^{(l)} = W^{(l)} a^{(l-1)} + b^{(l)}$$

$$a^{(l)} = \sigma\left(z^{(l)}\right)$$

$W^{(l)}$ are weights, $b^{(l)}$ are biases, $\sigma(\cdot)$ is the activation function, $a^{(l-1)}$ are the activations from the last layer. For the backward step, we have:

$$\delta^{(l)} = \frac{\partial \mathcal{L}}{\partial z^{(l)}} = \frac{\partial \mathcal{L}}{\partial a^{(l)}} \frac{\partial a^{(l)}}{\partial z^{(l)}} = \frac{\partial \mathcal{L}}{\partial a^{(l)}} \sigma'\left(z^{(l)}\right)$$

$$\delta^{(L)} = \frac{\partial \mathcal{L}}{\partial a^{(L)}} \sigma'\left(z^{(L)}\right)$$

$$\delta^{(l)} = \frac{\partial \mathcal{L}}{\partial z^{(l+1)}} \frac{\partial z^{(l+1)}}{\partial a^{(l)}} \sigma'\left(z^{(l)}\right) = \left(W^{(l+1)}\right)^T \delta^{(l+1)} \sigma'\left(z^{(l)}\right)$$

Backpropagation Recap: Chain Rule and Gradients

For a parameter $W^{(l)}$, the loss depends on it through:

$$W^{(l)} \to z^{(l)} \to a^{(l)} \to \cdots \to \mathcal{L}$$

Applying the chain rule:

$$\frac{\partial \mathcal{L}}{\partial W^{(l)}} = \frac{\partial \mathcal{L}}{\partial z^{(l)}} \frac{\partial z^{(l)}}{\partial W^{(l)}} = \delta^{(l)} \frac{\partial z^{(l)}}{\partial W^{(l)}} = \delta^{(l)} \left(a^{(l-1)}\right)^T$$

The parameter update is:

$$W_{t+1}^{(l)} \leftarrow W_t^{(l)} - \eta \frac{\partial \mathcal{L}}{\partial W_t^{(l)}}$$

$$W_{t+1} = W_t - \eta G$$

where η is the learning rate.

Optimizing the parameters of Neural Networks

Optimization problem: train a NN on a dataset D given a cost function L.

$$\min_W L(W) = \frac{1}{N} \sum_{i \in D} L_i(W)$$

Gradient Descent (full batch learning)

$$W_{t+1} = W_t - \eta_t g$$

$$g = \frac{1}{N} \sum_{i \in D} g_i$$

$$g_i = \nabla L_i(W_t)$$

$$W_{t+1} = W_t - \eta_t \frac{1}{N} \sum_{i \in D} \nabla L_i(W_t)$$

Optimizing the parameters of Neural Networks

Online Stochastic Gradient Descent

$$W_{t+1} = W_t - \eta_t \nabla L_{i_t}(W_t)$$

Mini-batch Stochastic Gradient Descent

$$W_{t+1} = W_t - \eta_t \hat{g}_t$$

$$\hat{g}_t = \frac{1}{BS} \sum_{i \in B_t} \nabla L_i(W_t), \quad \text{BS} = \text{batch size}$$

$$W_{t+1} = W_t - \eta_t \left(\frac{1}{BS} \sum_{i \in B_t} \nabla L_i(W_t) \right)$$

Optimizers - SGD

$$W_{t+1} \leftarrow W_t - \eta \hat{g}_t$$

```
def sgd(weights, grads, momentum_buffer_list, weight_decay,
    momentum, lr, nesterov):
    for i, weight in enumerate(weights):
        dW = grads[i]  # gradient for weight on layer i
        weight = weight - lr * dW
        weights[i] = weight
```

Optimizers - SGD + Momentum

$$M_t \leftarrow \mu M_{t-1} + \hat{g}_t$$

$$W_{t+1} \leftarrow W_t - \eta M_t$$

```
def sgd(weights, grads, momentum_buffer_list, weight_decay,
    momentum, lr, nesterov):
    for i, weight in enumerate(weights):
        dW = grads[i]  # gradient for weight on layer i
        if momentum != 0:
            buf = momentum_buffer_list[i]  # get momentum from t-1
            buf = buf * momentum + dW
            momentum_buffer_list[i] = buf  # update momentum buffer
            if nesterov:  # use Nesterov rule
                buf = dW + momentum * buf
            dW = buf  # the gradient becomes the momentum
        weight = weight - lr * dW
        weights[i] = weight
```

Optimizers - SGD + regularization

L2: $W_{t+1} \leftarrow W_t - \eta \hat{g}_t - \eta \lambda W_t$

```
def sgd(weights, grads, momentum_buffer_list, weight_decay,
    momentum, lr, nesterov):
    for i, weight in enumerate(weights):
        dW = grads[i]  # gradient for weight on layer i
        if weight_decay != 0:
            dW = dW + weight * weight_decay
        if momentum != 0:
            ...
        weight = weight - lr * dW
        weights[i] = weight
```

L1: $W_{t+1} \leftarrow W_t - \eta \hat{g}_t - \eta \lambda \, \text{sign}(W_t)$

Optimizers - Adam

Adam is one of the most popular optimizers.

$$M_t \leftarrow \beta_1 M_{t-1} + (1-\beta_1)\hat{g}_t, \quad \text{(from Momentum)}$$

$$v_t \leftarrow \beta_2 v_{t-1} + (1-\beta_2)\hat{g}_t^2, \quad \text{(from RMSProp)}$$

$$\hat{m}_t \leftarrow \frac{M_t}{1-\beta_1^t}, \qquad \hat{v}_t \leftarrow \frac{v_t}{1-\beta_2^t}$$

$$W_{t+1} \leftarrow W_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

AdamW:

$$W_{t+1} \leftarrow W_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \eta\lambda W_t$$

Optimizers - Muon

Muon is a recent state-of-the-art optimizer: link.

$$M_t \leftarrow \mu M_{t-1} + \hat{\mathrm{g}}_t$$

$$O_t \leftarrow \mathrm{NewtonSchulz5}(B_t)$$

$$W_{t+1} \leftarrow W_t - \eta O_t$$

Muon can be used to optimize only 2D weight matrices. For other parameters, AdamW is used instead.

Optimizers - other optimizers

- A visualization of different optimizers: link.

Learning Rate Schedulers

- $W_{t+1} \leftarrow W_t - \eta \hat{g}_t$
- bigger learning rates may cause overshooting;
- smaller learning rates lead to slow convergence.

Learning Rate Schedulers

- Illustration of learning rate policies: link.

Batch Size Schedulers

- $W_{t+1} \leftarrow W_t - \eta \hat{g}_t$
- Decreasing the learning rate is proportional to increasing the batch size: $W_{t+1} = W_t - \eta_t \left(\frac{1}{BS} \sum_{i \in B_t} \nabla L_i(W_t) \right)$
- Batch size scheduling can decrease the training time by more than 30% due to better hardware utilization.
- Batch size schedulers implementation: link.
- Visualization of batch size schedulers: link.

References

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